7.1.2024

Tema 05

<u>Ex. 2:</u>

$Z \sim N(0,1) \Rightarrow \mathbb{E}[Z] = 0$

$Var[Z] = 1$

$X := \sigma \cdot Z \Rightarrow \mathbb{E}[X] = \mathbb{E}[\sigma \cdot Z] = \sigma \cdot \mathbb{E}[Z] = 0$

$Var[X] = Var[\sigma \cdot Z] = \sigma^2 \cdot Var[Z] = \sigma^2$

Fie $a \in \mathbb{R}$

$$\mathbb{P}(X \leq a) = \mathbb{P}(\sigma \cdot Z \leq a) = \mathbb{P}\left(Z \leq \frac{a}{\sigma}\right) =$$

$$= \int_{-\infty}^{\frac{a}{\sigma}} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{z^2}{2}} dz \underset{\text{s.v.}}{=} \int_{-\infty \cdot \sigma}^{\frac{a}{\sigma} \cdot \sigma} \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{x^2}{2\sigma^2}} \cdot \frac{dx}{\sigma} =$$

$$\begin{cases} x = \sigma \cdot z \Rightarrow z = \frac{x}{\sigma} \\ dx = \sigma dz \Rightarrow dz = \frac{dx}{\sigma} \end{cases}$$

$$= \int_{-\infty}^{a} \frac{1}{\sqrt{2\pi \cdot \sigma^2}} \cdot e^{-\frac{x^2}{2\sigma^2}} dx = \int_{-\infty}^{a} p_X(x)\, dx \Rightarrow$$

$$\Rightarrow p_X(x) = \frac{1}{\sqrt{2\pi\, \sigma^2}} \cdot e^{-\frac{x^2}{2\sigma^2}} \Rightarrow X \sim \mathcal{N}(0, \sigma^2)$$

$$\mu = 0$$

<u>Ex. 3:</u>

$X \sim Exp(\lambda)$; $\alpha > 0$

$Y := \alpha \cdot X$

$Z : X^2$

$$p_X(t) = \begin{cases} \lambda e^{-\lambda t} & t \geq 0 \\ 0 & ; \text{rest} \end{cases}$$

Fie $a \in \mathbb{R}$.

$$\mathbb{P}(Y \leq a) = \mathbb{P}(\alpha \cdot X \leq a) = \mathbb{P}\left(X \leq \frac{a}{\alpha}\right) =$$

$$= \int_{-\infty}^{\frac{a}{\alpha}} p_X(x)\, dx = \underbrace{\int_{-\infty}^{0} p_X(x)\, dx}_{=0} + \int_{0}^{\frac{a}{\alpha}} p_X(x)\, dx =$$

$$= \int_{0}^{\frac{a}{\alpha}} \lambda \cdot e^{-\lambda x} dx \underset{\text{S.V.}}{=} \int_{0}^{\frac{a}{\alpha} \cdot \alpha} \lambda \cdot e^{-\lambda \cdot \frac{y}{\alpha}} \cdot \frac{dy}{\alpha} =$$

$$\begin{cases} y = \alpha \cdot x \Rightarrow x = \frac{y}{\alpha} \\ dy = \alpha \cdot dx \Rightarrow dx = \frac{dy}{\alpha} \end{cases}$$

$$= \int_0^a \frac{\lambda}{\alpha} \cdot e^{-\lambda \frac{y}{\alpha}} \cdot dy \Rightarrow$$

$$\Rightarrow p_Y(t) = \begin{cases} \frac{\lambda}{\alpha} \cdot e^{-\frac{\lambda}{\alpha} \cdot t} & \text{pt. } t \geq 0 \\ 0 \text{ ; rest.} \end{cases} \Bigg| \Rightarrow Y \sim Exp\left(\frac{\lambda}{\alpha}\right)$$

Fie $a \in \mathbb{R}$.

$$\mathbb{P}(Z \leq a) = \mathbb{P}(X^2 \leq a) = \mathbb{P}(X \leq \sqrt{a}) =$$

$$= \int_{-\infty}^{\sqrt{a}} p_X(x)\,dx = \underbrace{\int_{-\infty}^{0} p_X(x)\,dx}_{=0} + \int_0^{\sqrt{a}} p_X(x)\,dx =$$

$$= \int_0^{\sqrt{a}} p_X(x)\,dx = \int_0^{\sqrt{a}} \lambda e^{-\lambda x}\,dx =$$

S.V.

$$\begin{cases} z = x^2 \Rightarrow x = \sqrt{z} \\ dz = 2x\,dx \Rightarrow dx = \frac{dz}{2x} = \frac{dz}{2\sqrt{z}} \end{cases}$$

$$= \int_0^a \lambda \cdot e^{-\lambda \cdot \sqrt{z}} \cdot \frac{dz}{2\sqrt{z}} = \int_0^a \frac{\lambda}{2\sqrt{z}} \cdot e^{-\lambda\sqrt{z}}\,dz \Rightarrow$$

$$\Rightarrow p_Z(t) = \begin{cases} \frac{\lambda}{2\sqrt{t}} \cdot e^{-\lambda\sqrt{t}} & \text{; pt. } t \geq 0 \\ 0 & \text{pt. rest} \end{cases}$$

$p_Z(t) \sim \text{Weibull}\left(t, \frac{1}{\lambda^2}, \frac{1}{2}\right)$,

unde $\text{Weibull}(t; \lambda; k) = \begin{cases} \frac{k}{\lambda} \cdot \left(\frac{t}{\lambda}\right)^{k-1} e^{-\left(\frac{t}{\lambda}\right)^k}; \text{ pt. } t \geq 0 \\ 0; \text{ rest} \end{cases}$

<u>Ex. 4:</u>

$X \sim \text{Exp}(\lambda)$

Fie $Y := [X] + 1$ $\qquad Y: \Omega \to \mathbb{N}^* = \{1, 2, \dots\}$

Fie $n \in \mathbb{N}^*$

$$\mathbb{P}(Y = n) = \mathbb{P}([X] + 1 = n) = \mathbb{P}([X] = n-1) =$$

$$= \mathbb{P}(n-1 \leq X < n) = \int_{n-1}^{n} p_X(x)\,dx =$$

$$= \int_{n-1}^{n} \lambda e^{-\lambda x}\,dx = \int_{n-1}^{n} -\left(e^{-\lambda x}\right)'\,dx =$$

$$= -\int_{n-1}^{n} \left(e^{-\lambda x}\right)'\,dx = -\left[\left(e^{-\lambda x}\right)\Big|_{n-1}^{n}\right] =$$

$$= -e^{-\lambda n} + e^{-\lambda(n-1)} = e^{-\lambda(n-1)}\left(1 - e^{-\lambda}\right) =$$

$$= \left(e^{-\lambda}\right)^{n-1}\left(1 - e^{-\lambda}\right) = (1-p)^{n-1} \cdot p \quad \cancel{\Rightarrow [X]} \Rightarrow$$

Notăm $p = 1 - e^{-\lambda}$

$$\Rightarrow [X]+1 \sim Geom(1-e^{-\lambda})$$

pt. $n=1$ avem

$$\mathbb{P}(Y=1)=\mathbb{P}([X]=0)=\mathbb{P}(0\le X<1)=\int_0^1 \lambda e^{-\lambda x}dx=$$

$$=-\int_0^1 (e^{-\lambda x})'dx=-\left[(e^{-\lambda x})\Big|_0^1\right]=-e^{-\lambda}+e^0=1-e^{-\lambda}=$$

$$=(1-e^{-\lambda})(e^{-\lambda})^{n-1}$$

pt. $n=1$

<u>Ex. 5:</u>

$$f(x)=\begin{cases} K-\frac{x}{50}, & \text{pt. } 0\le x\le 10 \\ 0, & \text{altfel} \end{cases}$$

a) $$\int_{-\infty}^{\infty} f(x)dx=\underbrace{\int_{-\infty}^{0} f(x)dx}_{=0}+\int_0^{10} f(x)dx+$$

$$+\underbrace{\int_{10}^{+\infty} f(x)dx}_{=0}=\int_0^{10} f(x)dx=$$

$$=\int_0^{10}\left(K-\frac{x}{50}\right)dx=\left(Kx-\frac{x^2}{100}\right)\Big|_0^{10}=$$

$= 10K - 1 = 1 \Rightarrow 10K = 2$

$$\boxed{K = \frac{1}{5}}$$

b) $\mathbb{P}(X \leq 5) = \int_0^5 \left(\frac{1}{5} - \frac{x}{50}\right) dx =$

$= \left(\frac{x}{5} - \frac{x^2}{100}\right)\Big|_0^5 = 1 - \frac{25}{100} = \frac{3}{4} = 75\%$

c) $\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\, dx =$

$= \underbrace{\int_{-\infty}^{0} x f(x)\, dx}_{0} + \int_0^{10} x f(x)\, dx + \underbrace{\int_{10}^{+\infty} x f(x)\, dx}_{0} =$

$= \int_0^{10} x \cdot \left(\frac{1}{5} - \frac{x}{50}\right) dx = \int_0^{10} \left(\frac{x}{5} - \frac{x^2}{50}\right) dx =$

$= \left(\frac{x^2}{10} - \frac{x^3}{150}\right)\Big|_0^{10} = \frac{100}{10} - \frac{1000}{150} = 10 - \frac{100}{15} =$

$= \frac{150}{15} - \frac{100}{15} = \frac{50}{15} = \frac{10}{3} = 3,(3) \simeq 3,34$ ani

aprox. 3 ani 4 luni.

$\mathbb{E}[X^2] = \int_0^{10} x^2 \cdot \left(\frac{1}{5} - \frac{x}{50}\right) dx =$

$$= \int_0^{10} \left(\frac{x^2}{5} - \frac{x^3}{50} \right) dx = \left(\frac{x^3}{15} - \frac{x^4}{200} \right) \Bigg|_0^{10} =$$

$$= \frac{1000}{15} - \frac{10000}{200} = \frac{200}{3} - 50 = \frac{200-150}{3} = \frac{50}{3}$$

$$Var[X] = E[X^2] - E[X]^2 = \frac{50}{3} - \frac{100}{9} = \frac{50}{9} \simeq 5,(5)$$

$$\simeq 5,55$$